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Hyperspectral Raman Line Mapping as an Effective Tool To Monitor the Coating Thickness of Pharmaceutical Tablets

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Supporting Information

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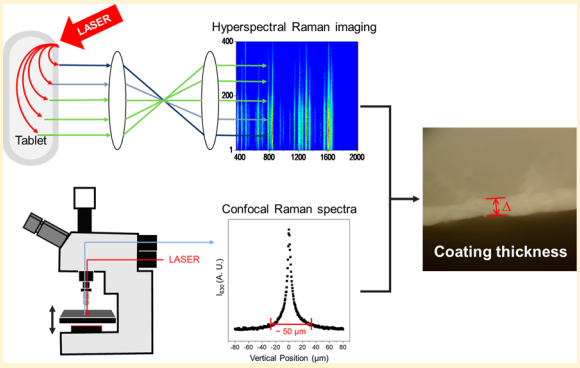
ABSTRACT: Protective chemical coatings are deposited on drugs during the manufacturing process for the purpose of controlling the pharmacokinetics of active pharmaceutical ingredients (APIs). Although manufacturers attempt to coat all the tablets uniformly, the film thickness of an individual drug is statistically different and depends on the measuring position of the anisotropic structure, and analytical methods for measuring coating thickness must be robust to statistical and geometrical aberrations. Herein, we demonstrate that a spatially offset Raman-spectroscopy-based line mapping method offered excellent calibration and prediction of the coating thickness of 270 acetaminophen (*N*-acetyl-para-aminophenol, paracetamol) tablets. Raman-scattered light resurfaced back from the coating and APIs, and offset-resolved spectra were projected according to the vertical positions in an imaging sensor. The Raman intensity ratio between the coating substance and the inner APIs is a key parameter in the analysis, and its variation with respect to the spatial offset is proportional to the coating thickness and duration. The results of this study have implications for the rapid spectroscopic thickness measurement of industrial products coated with transparent or translucent materials.



Vibrational spectroscopy has been extensively evaluated as a promising candidate for process analytical technology (PAT), thereby enabling fast and nondestructive quality evaluation in diverse pharmaceutical areas such as the measurement of tablet-coating thickness,[1−4] the determination of mixing homogeneity,[5−8] and verification of the concentrations of active pharmaceutical ingredients (APIs).[9−11] This is because it provides rich chemical and structural information on the constituents of a sample, and measurements with it are easy to expand for online analysis. In most of the applications, spectroscopic information obtained from a sample is directly indicative of target properties such as API concentration and crystallinity, and so, building multivariate or univariate models using the respective spectra to determine these properties is rather straightforward.

Among the analyses of pharmaceutical samples, those for measuring the coating thickness on a tablet such as Raman or near-infrared (NIR) spectroscopy have different characteristics, because the obtained peak intensities of the coated materials do not directly indicate absolute coating thickness and semiquantitatively vary according to coating thickness. Also, confocal laser scanning microscopy and optical coherence tomography can directly provide absolute coating thickness,[12,13] but their usage is limited to measurements of coating thickness in a microscopic area only and cannot be expanded for the other applications mentioned previously.

Research on the determination of coating thickness using Raman and NIR spectroscopy has been systematically studied by several groups.[14−16] For the determination, a calibration model is necessary to correlate the acquired spectral features with the coating thickness analyzed via a reference method. As summarized in the review papers, multivariate models using partial least-squares have mostly been employed to predict coating thickness, because the band of the coating material considerably overlaps those of the tablet constituents.[2,15] Although the measurements of coating thickness by both spectroscopic methods have been well-studied, a more robust method could provide more specific spectral information on coating thickness and employ a simple univariate calibration strategy rather than multivariate modeling, in which accuracy is sensitively degraded by sample morphology changes as well as

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instrumental variation. In this context, spatially offset Raman-spectroscopy (SORS)-based hyperspectral line mapping could have high potential to meet this demand.[17,18] In SORS measurements, the position of the Raman photon collection is away from that of the laser excitation, so when SORS line mapping is performed along a coated tablet, the ratios of the peak intensity between the outer coating material and the inner core tablet would vary in a series of mapped spectra. For example, the intensity of the coating peak is expected to decrease as the offset distance becomes longer. Moreover, the slope of the changes in intensity ratio is related to the coating thickness (i.e., a greater slope infers a thicker coating). This potential is the main motivation that drove this research. The overall concept of SORS line mapping to determine the coating thickness is described in the following section.

In the present study, a hyperspectral Raman system incorporating a large-area charge-coupled device (CCD) was used to quickly acquire SORS line mapping spectra simultaneously over a long distance. To initially examine the capability of the proposed scheme to measure the coating thickness, different numbers of transparent polyethylene terephthalate (PET) films were positioned above a packing of acetaminophen (*N*-acetyl-para-aminophenol, paracetamol) powder, and the corresponding line mapping spectra were acquired using laser illumination on the top of the PET films. Afterward, the slope of the intensity ratios between the PET and acetaminophen peaks was examined according to the total thicknesses of the films. Next, hyperspectral line mapping spectra of real coated tablets collected at different stages in the coating process were acquired using the same instrumentation. Subsequently, the slopes of the intensity ratios between the coating material (TiO_2) and the core tablet (acetaminophen) peaks were correlated with the coating duration, and the resulting accuracy was evaluated. Finally, the potential of the proposed SORS line mapping scheme is discussed and compared with that of the conventional weight-gain method.

Overall SORS Line Mapping Scheme To Determine the Coating Thickness. Figure 1 describes the overall scheme of SORS line mapping for the measurement of the coating (outer layer) thickness of a sample. In Figure 1(a), the blue and green rectangles on the left indicate the outer coatings and inner samples, respectively; the coating (blue rectangle) is thinner in the upper case. Upon laser illumination on the top of the outer layer, offset-dependent Raman signals of the samples are projected along vertical rows of the CCD (right side), and a series of offset-dependent spectral slices are obtained, as shown in Figure 1(b). The presented spectra are just conceptual and normalized to compensate for the decrease in the entire peak intensities with increasing offset. The blue and green bands in the marked boxes correspond to the Raman peaks of the outer layer and the inner sample, respectively, and are used in the subsequent analysis.

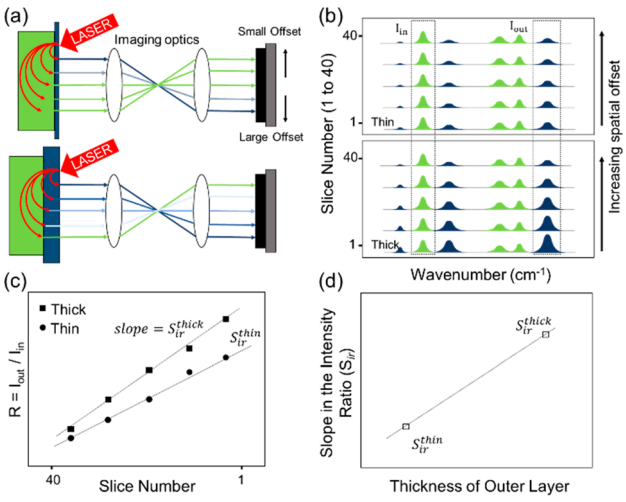

Figure 1. SORS line mapping scheme for the measurement of the pharmaceutical tablet-coating (outer layer) thickness: (a) a schematic of the optical configuration for acquiring SORS spectra, (b) a series of offset-dependent spectral slices collected by the scheme, (c) variations of I_{out}/I_{in} in the spectral slices in the cases of thin and thick coatings, and (d) a plot of the thickness of the outer layer vs S_{ir}.

It is apparent that the intensity of the inner layer (I_{in}) gradually decreases as the offset distance becomes longer, whereas the intensity of the outer layer (I_{out}) rapidly decreases under the same circumstances. Therefore, the intensity ratio between I_{out} and I_{in} (designated as I_{out}/I_{in}) varies continuously with varying (in this case, decreasing) offset distance, and the slope of the intensity ratio (S_{ir}) is dis-similar depending on the thickness of the outer layer, as shown in Figure 1(c). For example, I_{out}/I_{in} is highest at a zero offset and decreases with a longer offset distance, so the degree of decrease in I_{out}/I_{in} is thickness-dependent, and S_{ir} is determined using linear regression. Finally, at various coating thicknesses, the corresponding S_{ir} values are obtained, and a plot of the thickness of the outer layer vs S_{ir} is generated, as shown in Figure 1(d). The regression calculated in this plot is the final calibration curve for the determination of thickness. It is worthwhile to note that the proposed strategy works well when the outer layer is transparent or translucent.

■ EXPERIMENTAL SECTION

Tablet Samples. The biconvex tablets (height: 5 mm, diameter: 18 mm) employed in a previous publication[16] were used in this study (refer to the picture of a tablet in Figure S1). The components of the tablets were mainly acetaminophen, pregelatinized starch, croscarmellose sodium, and hydroxypropyl methyl cellulose. A total of 270 tablets with varying coating thicknesses were acquired from three separate batches (designated as B1, B2, and B3) prepared via a real coating process. The tablets were sampled 7, 8, and 12 times in B1, B2, and B3, respectively, during the 7 h coating period, and 10 tablets were collected at each sampling point. Tablet weight was measured using a high-precision balance (XS205 Dual Range, Mettler Toledo, OH, USA). The coating material was OPADRY WHITE (Colorcon, PA, USA), containing 62.50% hydroxypropyl methyl cellulose, 31.25% TiO_2, and 6.25% polyethylene glycol 400.

Instrumentation for the SORS Line Mapping. Raman hyperspectral images were acquired using an in-house wide-depth SORS system, which is described elsewhere in detail.[18] It consisted of a polychromator (Acton SP2300, Princeton Instruments, NJ, USA), a large-area CCD camera (PIXIS 400BR, Princeton Instruments, NJ, USA), a 785 nm diode laser (LML-785.0, PD-LD, NJ, USA), and relevant optical components. For spectral acquisition, the laser (power: 100 mW) illuminated the top of a tablet for 3 s, and Raman photons resurfacing back below the laser spot were collected via a couple of achromatic lenses. Next, the photons were allowed to pass through a long-pass filter (LL01-785-25, Semrock, NY, USA) and were projected onto the slit of the

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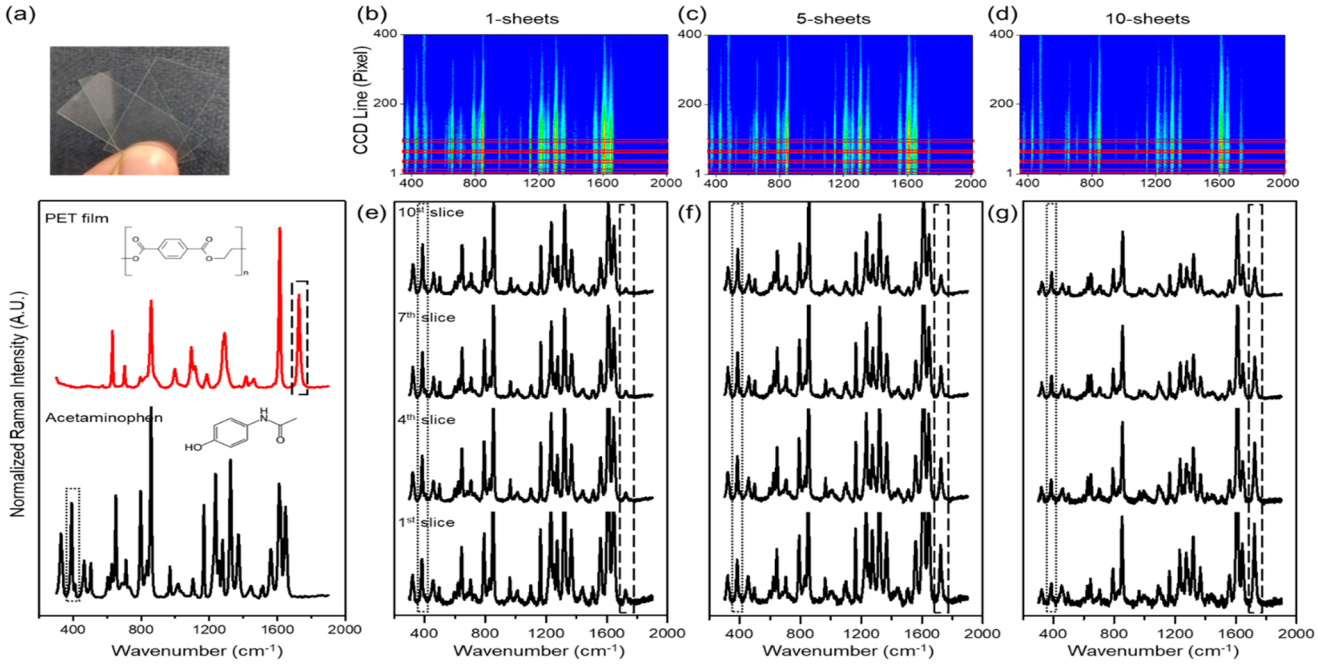

Figure 2. (a) Raman spectra of paracetamol and PET film and the photograph of the PET films. (b−d) Raman hyperspectral images of the model sample when the number of PET films on top of the paracetamol powder was 1, 5, and 10, and (e−g) four selected (1st, 4th, 7th, and 10th) spectral slices when the number of PET films was 1, 5, and 10. The dotted and dashed lines indicate the paracetamol peak at 390 cm^{-1} and the PET peak at 1728 cm^{-1}, respectively.

polychromator. Finally, offset-dependent Raman photons were dispersed by grating and imaged onto a CCD with 1340 × 400 pixels. And the calculated spatial offset for one vertical pixel corresponds to 25 μm.

A confocal study was performed using lab-built instrumentation comprising an upright microscope (Olympus, BX53), a 532 nm DPSS laser (532S-50-COL-PP, Oxxius, Lannion, France), a polychromator (Shamrock 303i, Andor, Belfast, Northern Ireland), and a CCD (iDUS-DU-420A-BEX2-DD, Andor, Belfast, Northern Ireland). The sample position was controlled with an *XYZ* motorized stage (LEE200, ST1, Seoul, Republic of Korea) providing an axial resolution of 0.10 μm. The calculated axial optical resolution was 1.18 μm, which was obtained by considering the numerical aperture (NA = 0.9) of a 100× objective lens (MPLFLN, Olympus, Tokyo, Japan).

After the collection of the raw Raman spectra, baseline corrections were performed using asymmetric least-squares (AsLS).[19] Peak intensity ratios were calculated using Matlab 2018 (MathWorks, MA, USA), and OriginPro 9.0 (OriginLab, MA, USA) software was used to correlate the intensity ratio with coating duration.

■ RESULTS AND DISCUSSION

Evaluation of the SORS Line Mapping Scheme Using a Model Sample. Before the analysis of real coated pharmaceutical tablets, the potential of the proposed scheme was examined using a well-defined sample: acetaminophen powder covered with PET film, which can be regarded as a core tablet and an outer coating, respectively. Because PET film is optically transparent and its thickness can be accurately measured, the model sample was sufficiently versatile for a basic evaluation of the scheme. The thickness of a single PET sheet was 140 μm, and multiple sheets of the film were stacked to increase the overall thickness to emulate an increase in coating thickness. Figure 2(a) shows Raman spectra of the acetaminophen and the PET film: the baseline-corrected spectra using AsLS are presented here, and the picture of the PET films is also shown. Major Raman peaks of the PET film at 886 (O−CH_2 and C−C stretches of the gauche ethylene glycol unit), 1613 (ring C1−C4 stretching), and 1728 cm^{-1} (C = O stretching) were apparent,[20] and for acetaminophen, many diverse peaks corresponding to its molecular structure appeared over the entire spectral range. The acetaminophen peak at 390 cm^{-1} and the PET peak at 1728 cm^{-1} (marked with dotted and dashed lines, respectively) were selected for further analysis, because these two bands did not overlap with any of the others.

Figure 2(b−d) shows Raman hyperspectral images of the model sample when the number of PET films on top of the acetaminophen powder was 1, 5, and 10, respectively. The vertical dimension of the CCD was 8 mm, so the magnification was adjusted to drive the hyperspectral SORS image to fully cover it by changing the position of the two lenses. As can be seen, the intensities around the first row of the CCD corresponding to Raman signals generated at or near the location of the laser illumination were strong and became substantially weaker around the 200th row, which is equivalent to the collection of photons located much farther away from the laser illumination. There are two factors resulting in the observed trend. First, as the offset distance became longer, the laser photons penetrated deeper into the sample, and the Raman photons were accordingly generated farther away from the sample surface. Because the scattering direction of the incident photons was randomized after multiple scattering events, and the photon distribution at the longer offset distance dwindled, the number of photons reaching the CCD decreased under this condition, consequentially decreasing the peak intensity. Second, the collection efficiency was highest when light passes through the center of the concave mirror and diffraction grating in the polychromator, whereas it decreased

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toward both edges. Hence, the peak intensities at the first row were apparently weaker than at the 50th row even when the laser was focused on the former's position.

As mentioned previously, the CCD with 400 rows offered 400 offset-dependent spectra, whereas 10 consecutive spectra (such as the 1st through 10th spectra) were averaged into one spectral slice to improve the signal-to-noise ratio. Therefore, a total of 40 spectral slices along the offset were obtained for each sample and used in the analysis. Figure 2(e−g) shows four selected (1st, 4th, 7th, and 10th) spectral slices when the number of PET films in the measurement was 1, 5, and 10, respectively. Note that each spectrum was normalized to the height of the nonoverlapped acetaminophen peak (390 cm^{-1}). The first spectral slice corresponds to the average of the 1st to the 10th spectral row containing information pertinent at the nearest point to the laser illumination. The dotted and dashed box indicate the acetaminophen peak at 390 and the PET peak at 1728 cm^{-1}, respectively. One clear observation is that the intensity of the PET peak became greater with increasing PET thickness, and the intensity of the acetaminophen peak decreased concurrently. In addition, at a given PET thickness, the intensity of the PET peak decreased rapidly from the 1st to the 10th spectral slice. As the offset distance became longer, the scattering of the photons from the acetaminophen underneath the PET film became more sampled, and a relative increase in the acetaminophen peak (with a simultaneous decrease in the PET peak) was the result. Therefore, the trend in the change in intensity ratio between both peaks could be related to the PET thickness.

Figure 3(a) shows $I_{PET}/I_{Acetaminophen}$ ($=I_{1728}/I_{390}$) values calculated from all 40 spectral slices acquired by varying the

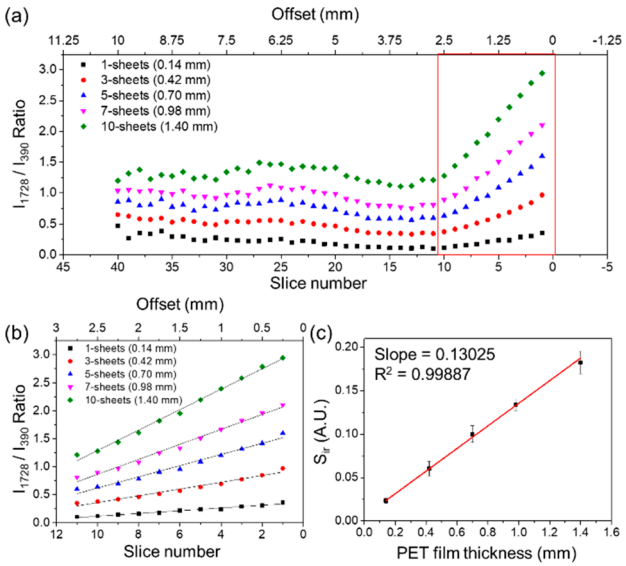

Figure 3. (a) I_{1728}/I_{390} ratios calculated from all 40 spectral slices acquired by varying the PET thicknesses from 0.14−1.40 mm (from 1−10 films), (b) the I_{1728}/I_{390} ratios obtained from only the 1st to the 11th spectral slice, and (c) the relationship between PET film thickness and S_{ir}.

PET thicknesses from 0.14−1.40 mm (from 1−10 films). In each case, the I_{1728}/I_{390} value at the first slice was the greatest, informing us of the dominance of the peak due to Raman scattering by the PET surface. The ratios became gradually smaller as the slice number increased (a longer offset distance) due to the greater sampling of the acetaminophen. In all cases, the decrease in the intensity ratio was the apparent tendency from the 1st to the 11th slice, whereas no systematic variations were observed below the 11th slice, as the photon density became smaller due to an increase in scattering by the dense powder medium and the sampled area at each slice being finite. Hence, the obtained spectral information would not have been sufficiently representative of the PET thickness variation, which would have lowered the signal-to-noise ratio. Therefore, using the ratios for the 1st to the 11th spectral slices was desirable.

Figure 3(b) highlights I_{1728}/I_{390} values obtained from the 1st to the 11th spectral slices only. As shown, the slope in the intensity ratio (S_{ir}) was greatest at the PET thickness of 1.40 mm and decreases as the PET film became thinner (the slope in each case was individually obtained by linear regression), and it became very clear that S_{ir} is thickness-dependent. Figure 3(c) shows the relationship between PET thickness and S_{ir}: the latter rose linearly according to the variation in thickness, resulting in an R^2 of 0.998. The error bars in each case were obtained based on five independent measurements of the sample. The overall results confirm that the thickness of the outer layer of the PET−acetaminophen model sample could be measured using the SORS line mapping scheme.

Figure 4(a) shows Raman spectra of TiO_2 (a major coating material in the coating spray) and a core (noncoated) tablet.

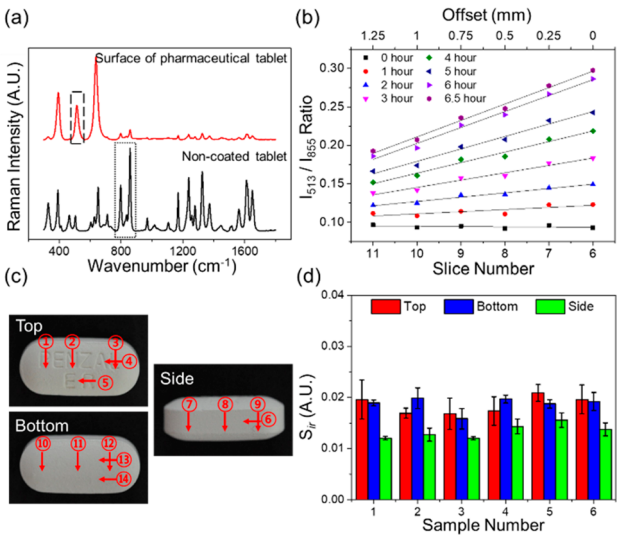

Figure 4. (a) Raman spectra of TiO_2 and a core tablet, (b) I_{TiO2}/I_{Core} values calculated using the 6th to the 11th spectral slices collected from eight tablets acquired during the coating time from 0 to 6.5 h, (c) top-, side-, and bottom-view photos of a test tablet (the arrows with the numbers indicate the locations of the SORS line mappings on the tablet), and (d) the average values of S_{ir} for the top-, bottom-, and side-face measurements.

As can be seen, the TiO_2 peaks were considerably intense, because the strong Raman scattering is known. For the calculation of the intensity ratio (I_{TiO2}/I_{Core}), the most intense peak at 855 cm^{-1} was selected along with the TiO_2 peak at 513 cm^{-1}, because it overlapped the least with the peaks of the core tablet. Considering that the spectral overlap of each material was limited to the low-energy region, we extended the selection bandwidth of the core material to improve the signal-to-noise ratio. In addition, we used another peak at 638

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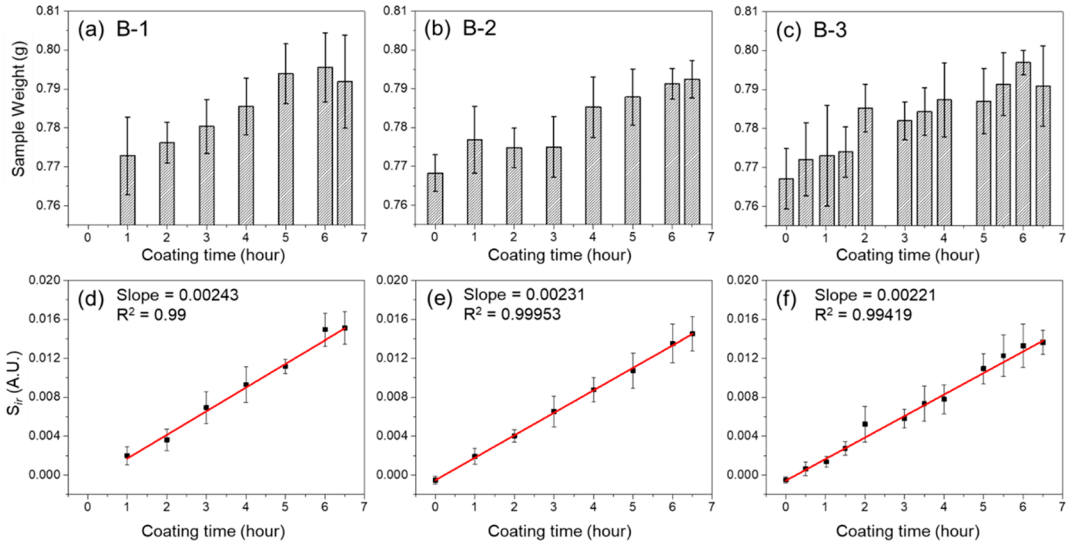

Figure 5. (a−c) Variations of the average tablet weight during the 7 h coating period for B1, B2, and B3 samples, and (d−f) the correlations between the coating time and the corresponding S_{ir} values for B1, B2, and B3.

instead of 513 cm^{-1} and showed the results for comparison (in Figure S6).

Initially, a range of spectral slices providing linear variation in I_{513}/I_{855} was sought, because this would be different from the model sample case. For the evaluation, eight tablets acquired during coating times from 0 to 6.5 h (the B2 samples) were selected, and the bottom face of each tablet was measured. Subsequently, based on an examination of the I_{513}/I_{855} values calculated from all 40 spectral slices, the most distinct linear variation was observed from the 6th to the 11th spectral slice, as shown in Figure 4(b). Similar to the result using the model sample, S_{ir} became steeper as the coating time increased (resulting in a greater coating thickness).

Next, the reproducibility of the SORS line mapping with changing the measurement location in a tablet was evaluated. As shown, the letters indicating the commercial name of the tablet were engraved on the top with none on the bottom, whereas the surfaces of the tablet sides were noncurved without an engraving. For the evaluation, 14 different line mapping locations were selected, as indicated in the picture of the tablet (Figure 4(c)). The tail of the arrow is the location of the laser illumination, and SORS spectra were obtained in the tail-to-head direction. On the top face, five different locations were selected, with four of them crossing the engraved letters either vertically or horizontally. On the side face, vertical line mappings were performed at three different locations, and horizontal line mapping was executed at one location. On the bottom face, five different locations were either vertically or horizontally line-mapped.

The previously described line mapping was performed on one tablet collected at the coating time of 6.5 h. (Figure S5) For each location, the I_{TiO2}/I_{Core} value was calculated using the 6th to the 11th spectral slices, and then, the S_{ir} values were obtained. The average S_{ir} values for the line mappings on the top, bottom, and side faces were 0.0185 (13.7%), 0.0187 (7.3%), and 0.0134 (7.3%), respectively. The numbers in the parentheses indicate the corresponding relative standard deviations (RSDs). The variation was larger for the top-face measurements, because the mapped surface was more irregular due to the letter engraving; thus, these line mappings to determine coating thickness were not useful. On the other hand, the RSDs in the bottom- and side-face measurements were the same.

To determine which tablet face was suitable for reliable measurements, an additional five tablets (at the same coating time of 6.5 h) were measured, and the magnitudes of the S_{ir} values were compared. Figure 4(d) shows the S_{ir} averages for the top-, bottom-, and side-face measurements. The S_{ir} values for the top- and bottom-face measurements were higher and similar to each other, but the magnitudes of the error bars were greater for the former, as was also found earlier. Meanwhile, the S_{ir} values were smaller for the side-face measurements, indicating less sensitivity when determining the coating thickness. It is reasonable to speculate that the coating thickness on the side was relatively thinner than on the top or bottom because of the smaller area and less chance of exposure to the coating spray. When considering the overall results, the bottom-face measurement provided balanced reproducibility and sensitivity and thus was used in the rest of the study.

To evaluate the potential of the proposed scheme for a quantitative analysis, 270 tablets acquired from three separate batches (B1, B2, and B3) of a real coating process were used (refer to the Experimental Section). Figure 5(a−c) shows the variation in average tablet weights during the 7 h coating period for the B1, B2, and B3 samples, respectively. As mentioned, 10 tablets were sampled at a time. The error bars indicate the corresponding standard deviations. Because the RSD in the weights of the core (uncoated) tablets was 1.02%, either the same or greater weight variation was expected for the coated tablets afterward. As can be seen, the coating weights increased with the lengthening of the coating time for all three batches, and the increases were generally linear. The magnitudes of the error bars varied moderately during the coating process. Estimation of the coating thickness via the weight-gain method was simple and applicable even though there were limitations such as batch-to-batch weight variation in the core tablets and imperfect linearity, as shown in the plots.

Figure 5(d−f) shows the correlations between the coating time and the corresponding S_{ir} value for B1, B2, and B3, respectively, in which it is obvious that the coating thickness increased with increasing coating time. The regression results

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were a better linear fit compared to those for the tablet weights (Figure 5(a)−(c)), and the resulting R^2 values for all three batches were over 0.99. The slopes for B1, B2, and B3 were 0.00243, 0.00231, and 0.00221, respectively. On the basis of these results, the SORS line mapping approach is potentially more accurate and versatile for determining the coating thickness than the conventional weight-gain method. For example, these criteria for the tablets sampled at a coating time of 6.5 h from B1 and B3 were more accurately correlated with the coating time using the SORS line mapping.

To demonstrate the practical applicability of our method, it was necessary to evaluate the prediction performance. For this purpose, the B2 samples were assigned as the calibration set, and the obtained regression parameters were used to predict the coating times of the B1 and B3 samples. The prediction results (actual vs predicted coating time) are shown in Figure 6(a). As is evident, the relationship is apparently linear with

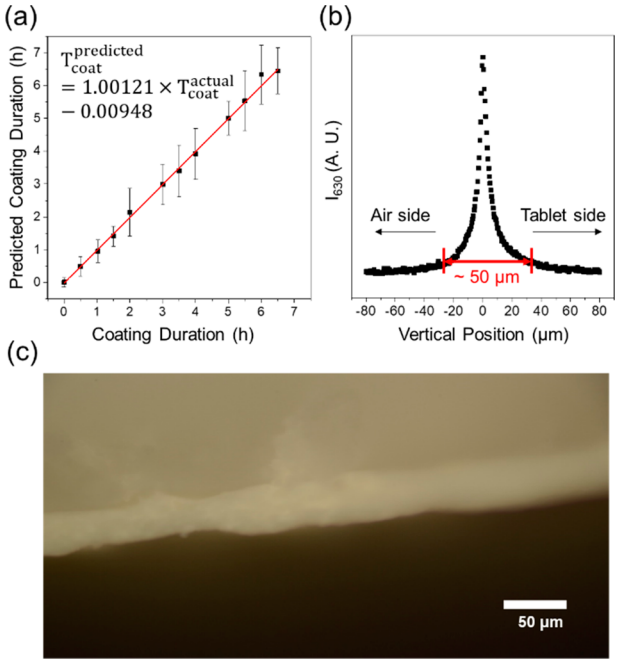

Figure 6. (a) Prediction result (actual vs predicted coating time) for the B1 and B3 samples, (b) intensity variation of the TiO_2 peak at 630 cm^{-1} in the confocal Raman measurement of a 6.5 h-coated tablet, and (c) optical image of cross section of the same tablet after cleaving.

good correlation, and the slope of prediction was nearly 1.0, indicating good accuracy. Meanwhile, the magnitudes of the error bars became larger from 2−7 h of coating, which is attributed to the increased tablet-to-tablet variation in the coatings.

The linear relationship between S_{ir} and coating time confirms that the proposed method could determine the coating thickness if absolute thickness is accurately known at selected coating durations. As a trial, confocal Raman microscopy was employed in this study. A 6.5 h-coated tablet was moved vertically with an objective lens fixed in an upright microscope, and Raman photons were transferred to a detector through a 105 μm optical fiber; the fiber core also served as a small pinhole, and the axial resolution of the microscope system was estimated as 1.18 μm. (Figure S2) The intensities of the coating (TiO_2) peak centered at 630 cm^{-1} were examined by continuously moving the sample vertically, the results of which are shown in Figure 6(b). A negative and positive position in the x-axis corresponds to the laser focusing on the outside and inside of the tablet, respectively, and the data were not fitted using typical Gaussian or Lorentzian functions. (A 50 μm scale is presented in the figure.) Additionally, we cleaved a tablet to measure a representative value of the coating thickness with a microscope. A 6.5 h-coated tablet produced in batch B2 was selected and cleaved with a blade. A cross section of the cleaved tablet was taken by photographing with the microscope, as shown in Figure 6(c). On the basis of our observations of the data, the coating thickness was estimated as ∼50 μm. According to the results of the same measurements accomplished for several other samples, the thicknesses somewhat varied depending on the tablet as well as the measurement location. Thus, the average coating thickness of a tablet will be required for the SORS line mapping attempts to directly determine the coating thickness in the future.

■ CONCLUSIONS

In this study, SORS line mapping was used to measure the physicochemical properties of the pharmaceutical coating as well as the inner tablet. We correlated the thickness of the coating substances with a slope of the Raman intensity ratio between the outer coating and the inner content and implemented a univariate model to predict the coating thickness. This method is distinguishable from other thickness-measurement methods in that it provides statistical information over the macroscopic (∼mm)-length scale, and the key feature of our instrument is obtaining multiple offset Raman spectra in a single hyperspectral plane. Although further improvement is required toward robustness to structural deviation, such as that caused by engraving and curvature, we expect that our hyperspectral Raman line mapping method could lead to the development of online PATs to perform both physical and chemical inspections of products protected by thin films at the same time.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.9b00047.

Image of an in-house wide-depth spatially offset Raman-spectroscopy (W-SORS) system; image of paracetamol tablets analyzed in the experiment; schematic and image of a lab-built confocal Raman microscope and the theoretical axial resolution of the confocal microscope in the experiment; plot of I_{1728}/I_{390} values according to vertical pixels for a PET film model without binning and for the linear response region of the 1st to 100th pixels; plot of I_{513}/I_{855} values with slice numbers for TiO_2-coated acetaminophen tablets with varying coating durations; slice spectra of an acetaminophen tablet (P-2, T_{coat} = 6.5) obtained at large offsets; plot of I_{638}/I_{855} values with slice numbers for a TiO2-coated acetaminophen tablet; plot of I_{638}/I_{855} values for the linear response region (5th to 10th slice); correlations between the coating time and the corresponding S_{ir} values (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Reich, G. *Adv. Drug Delivery Rev.* **2005**, *57* (8), 1109−1143.

(2) Andersson, M.; Folestad, S.; Gottfries, J.; Johansson, M. O.; Josefson, M.; Wahlund, K.-G. *Anal. Chem.* **2000**, *72* (9), 2099−2108.

(3) Kelley, W. P.; Chen, S.; Floyd, P. D.; Hu, P.; Kapsi, S. G.; Kord, A. S.; Sun, M.; Vogt, F. G. *Anal. Chem.* **2012**, *84* (10), 4357−4372.

(4) De Beer, T.; Burggraeve, A.; Fonteyne, M.; Saerens, L.; Remon, J. P.; Vervaet, C. *Int. J. Pharm.* **2011**, *417* (1−2), 32−47.

(5) Sekulic, S. S.; Ward, H. W.; Brannegan, D. R.; Stanley, E. D.; Evans, C. L.; Sciavolino, S. T.; Hailey, P. A.; Aldridge, P. K. *Anal. Chem.* **1996**, *68* (3), 509−513.

(6) De Beer, T. R. M.; Baeyens, W. R. G.; Ouyang, J.; Vervaet, C.; Remon, J. P. *Analyst* **2006**, *131* (10), 1137.

(7) Shin, K.; Chung, H. *Analyst* **2013**, *138* (12), 3335.

(8) Gordon, K. C.; McGoverin, C. M. *Int. J. Pharm.* **2011**, *417* (1−2), 151−162.

(9) Eliasson, C.; Macleod, N. A.; Jayes, L. C.; Clarke, F. C.; Hammond, S. V.; Smith, M. R.; Matousek, P. *J. Pharm. Biomed. Anal.* **2008**, *47* (2), 221−229.

(10) Park, S. C.; Kim, M.; Noh, J.; Chung, H.; Woo, Y.; Lee, J.; Kemper, M. S. *Anal. Chim. Acta* **2007**, *593* (1), 46−53.

(11) Buckley, K.; Matousek, P. *J. Pharm. Biomed. Anal.* **2011**, *55* (4), 645−652.

(12) Zhong, S.; Shen, Y.-C.; Ho, L.; May, R. K.; Zeitler, J. A.; Evans, M.; Taday, P. F.; Pepper, M.; Rades, T.; Gordon, K. C.; et al. *Opt. Lasers Eng.* **2011**, *49* (3), 361−365.

(13) Pygall, S. R.; Whetstone, J.; Timmins, P.; Melia, C. D. *Adv. Drug Delivery Rev.* **2007**, *59* (14), 1434−1452.

(14) Kirsch, J. D.; Drennen, J. K. *J. Pharm. Biomed. Anal.* **1995**, *13* (10), 1273−1281.

(15) Romero-Torres, S.; Pérez-Ramos, J. D.; Morris, K. R.; Grant, E. R. *J. Pharm. Biomed. Anal.* **2006**, *41* (3), 811−819.

(16) Kim, J.; Hwang, J.; Woo, Y.-A.; Chung, H. *J. Pharm. Biomed. Anal.* **2016**, *131*, 281−286.

(17) Matousek, P.; Parker, A. W. *Appl. Spectrosc.* **2006**, *60* (12), 1353−1357.

(18) Cho, Y.; Song, S. W.; Sung, J.; Jeong, Y.-S.; Park, C. R.; Kim, H. M. *Analyst* **2017**, *142* (19), 3613−3619.

(19) Zhang, Z.-M.; Chen, S.; Liang, Y.-Z. *Analyst* **2010**, *135* (5), 1138.

(20) Boerio, F. J.; Bahl, S. K.; McGraw, G. E. *J. Polym. Sci., Polym. Phys. Ed.* **1976**, *14* (6), 1029−1046.

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